

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285227

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

GOOSEN; Maikel Robert

SYSTEM AND METHOD FOR IMPROVING IMAGE QUALITY DURING INSPECTION

Abstract

Systems, apparatuses, and methods for improving image quality. In some embodiments, a method may include obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate of the plurality of images; and generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined maximum likelihood estimate.

Inventors: GOOSEN; Maikel Robert (Eindhoven, NL)

Applicant: ASML Netherlands B.V. (Veldhoven, NL)

Family ID: 81392730

Assignee: ASML Netherlands B.V. (Veldhoven, NL)

Appl. No.: 18/859954

Filed (or PCT Filed): March 28, 2023

PCT No.: PCT/EP2023/057947

Foreign Application Priority Data

EP 22170392.9 Apr. 27, 2022

Publication Classification

Int. Cl.: G06T5/50 (20060101); G06T5/73 (20240101); G06T7/00 (20170101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority of EP application Ser. No. 22/170,392.9 which was filed on 27 Apr. 2022 and which is incorporated herein in its entirety by reference.

FIELD

[0002] The description herein relates to the field of inspection systems, and more particularly to systems for improving image quality during inspection.

BACKGROUND

[0003] In manufacturing processes of integrated circuits (ICs), unfinished or finished circuit components are inspected to ensure that they are manufactured according to design and are free of defects. An inspection system utilizing an optical microscope typically has resolution down to a few hundred nanometers; and the resolution is limited by the wavelength of light. As the physical sizes of IC components continue to reduce down to sub-100 or even sub-10 nanometers, inspection systems capable of higher resolution than those utilizing optical microscopes are needed.

[0004] A charged particle (e.g., electron) beam microscope, such as a scanning electron microscope (SEM) or a transmission electron microscope (TEM), capable of resolution down to less than a nanometer, serves as a practicable tool for inspecting IC components having a feature size that is sub-100 nanometers. With a SEM, electrons of a single primary electron beam, or electrons of a plurality of primary electron beams, can be focused at locations of interest of a wafer under inspection. The primary electrons interact with the wafer and may be backscattered or may cause the wafer to emit secondary electrons. The intensity of the electron beams comprising the backscattered electrons and the secondary electrons may vary based on the properties of the internal and external structures of the wafer, and thereby may indicate whether the wafer has defects.

SUMMARY

[0005] Embodiments of the present disclosure provide apparatuses, systems, and methods for improving image quality. In some embodiments, systems, methods, and non-transitory computer readable mediums may include steps of obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate of the plurality of images; and generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined maximum likelihood estimate.

[0006] In some embodiments, systems, methods, and non-transitory computer readable mediums may include steps of obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate of the plurality of images; and generating a focus-adjusted image of the area based on the determined plurality of focus-related values and the determined maximum likelihood estimate.

[0007] In some embodiments, systems, methods, and non-transitory computer readable mediums may include steps of obtaining a plurality of images of an area in a field of view of a sample, wherein a first image of the plurality of images has a first focus-related value and a second image

of the plurality of images has a second focus-related value that is different from the first focus-related value; and generating a focus-adjusted image of the area in the field of view using the first and second images.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic diagram illustrating an exemplary electron beam inspection (EBI) system, consistent with embodiments of the present disclosure.

[0009] FIG. 2 is a schematic diagram illustrating an exemplary multi-beam system that is part of the exemplary charged particle beam inspection system of FIG. 1, consistent with embodiments of the present disclosure.

[0010] FIG. 3 is a schematic diagram of an exemplary system for improving image quality, consistent with embodiments of the present disclosure.

[0011] FIG. 4 is a schematic diagram illustrating an exemplary sample and an image generation schematic, consistent with embodiments of the present disclosure.

[0012] FIG. 5 is a flowchart illustrating an exemplary process of improving image quality, consistent with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0013] Reference will now be made in detail to exemplary embodiments, examples of which are illustrated in the accompanying drawings. The following description refers to the accompanying drawings in which the same numbers in different drawings represent the same or similar elements unless otherwise represented. The implementations set forth in the following description of exemplary embodiments do not represent all implementations consistent with the disclosure. Instead, they are merely examples of apparatuses and methods consistent with aspects related to the subject matter recited in the appended claims. For example, although some embodiments are described in the context of utilizing electron beams, the disclosure is not so limited. Other types of charged particle beams may be similarly applied. Furthermore, other imaging systems may be used, such as optical imaging, photodetection, x-ray detection, extreme ultraviolet inspection, deep ultraviolet inspection, or the like, in which they generate corresponding types of images.

[0014] Electronic devices are constructed of circuits formed on a piece of silicon called a substrate. Many circuits may be formed together on the same piece of silicon and are called integrated circuits or ICs. The size of these circuits has decreased dramatically so that many more of them can fit on the substrate. For example, an IC chip in a smart phone can be as small as a thumbnail and yet may include over 2 billion transistors, the size of each transistor being less than 1/1000th the size of a human hair.

[0015] Making these extremely small ICs is a complex, time-consuming, and expensive process, often involving hundreds of individual steps. Errors in even one step have the potential to result in defects in the finished IC rendering it useless. Thus, one goal of the manufacturing process is to avoid such defects to maximize the number of functional ICs made in the process, that is, to improve the overall yield of the process.

[0016] One component of improving yield is monitoring the chip making process to ensure that it is producing a sufficient number of functional integrated circuits. One way to monitor the process is to inspect the chip circuit structures at various stages of their formation. Inspection may be carried out using a scanning electron microscope (SEM). A SEM can be used to image these extremely small structures, in effect, taking a “picture” of the structures of the wafer. The image can be used to determine if the structure was formed properly and also if it was formed at the proper location. If the structure is defective, then the process can be adjusted so the defect is less likely to recur. Defects may be generated during various stages of semiconductor processing. For

the reason stated above, it is important to find defects accurately and efficiently as early as possible.

[0017] The working principle of a SEM is similar to a camera. A camera takes a picture by receiving and recording brightness and colors of light reflected or emitted from people or objects. A SEM takes a “picture” by receiving and recording energies or quantities of electrons reflected or emitted from the structures. Before taking such a “picture,” an electron beam may be provided onto the structures, and when the electrons are reflected or emitted (“exiting”) from the structures, a detector of the SEM may receive and record the energies or quantities of those electrons to generate an image. To take such a “picture,” some SEMs use a single electron beam (referred to as a “single-beam SEM”), while some SEMs use multiple electron beams (referred to as a “multi-beam SEM”) to take multiple “pictures” of the wafer. By using multiple electron beams, the SEM may provide more electron beams onto the structures for obtaining these multiple “pictures,” resulting in more electrons exiting from the structures. Accordingly, the detector may receive more exiting electrons simultaneously, and generate images of the structures of the wafer with a higher efficiency and a faster speed.

[0018] During inspection, it is advantageous to generate images (e.g., SEM images, optical images, x-ray images, photon images, etc.) having higher resolutions or sharpness so that the features (e.g., a contact, a metal line, a gate, etc.) on a sample in the images accurately represent the actual sample. In order to generate higher resolution or sharpness images, images of the features on the sample need to be in focus. Aberrations, such as a defocused electron beam, may result in blurred, low-quality images. To facilitate obtaining a high quality focus, the settings of an inspection system (e.g., the voltage or strength of an objective lens) may be adjusted to adjust the probe size of an electron beam.

[0019] In typical inspection systems, a focus measurement of the inspection system is performed to facilitate generating high quality images. A typical focus measurement involves obtaining images of a sample in an area outside of a Field of View (FOV). For example, the FOV may include areas of the sample to be inspected while the typical focus measurement obtains images of the sample outside of the areas to be inspected. The typical focus measurement involves obtaining a plurality of images of a sample outside of the FOV at different defocus values. For example, the objective lens may be adjusted (e.g., using different voltage values or different current values) before each image of the area outside of the FOV is obtained so that each image of the focus measurement has a different defocus value. The inspection setting (e.g., the voltage value of the objective lens) that corresponds to the highest resolution image or the highest sharpness image (e.g., determined by a key performance indicator (KPI) of resolution or sharpness) of the area outside of the FOV is used to obtain images of the sample within the FOV (e.g., one or more areas within the FOV, the entire FOV, etc.) during inspection.

[0020] In typical inspection systems, a plurality of images of the sample in the FOV are obtained using the inspection setting that corresponds to the highest resolution image or the highest sharpness image of the area outside of the FOV during the focus measurement. An average of the plurality of images (e.g., also referred to as a plurality of frames) of the sample within the FOV is determined to generate an inspection image of the sample in the FOV.

[0021] Typical focus measurements and inspections, however, suffer from constraints. An example of a constraint with typical inspections is that inspection throughput is low because multiple images are obtained for the focus measurement outside of the FOV and multiple images are obtained for the inspection measurement within the FOV. Another example of a constraint with typical inspections is that focus deviations may occur over a sample (e.g., the field of curvature may vary across the FOV), thereby reducing the accuracy and the quality of the generated inspection image since a focus measurement of an area outside of the FOV may not be applicable to an inspection image obtained from an area inside the FOV. Yet another example of a constraint with typical inspections is that noise within the obtained images may reduce the throughput of the generated

inspection images.

[0022] Some of the disclosed embodiments provide systems and methods that address some or all of these disadvantages by improving image quality during inspection. The disclosed embodiments may use a phase diversity analysis to determine focus-related values (e.g., defocus values) and a maximum likelihood estimate of images of a sample within a FOV, thereby enabling generation of a focus-adjusted (e.g., focus-corrected) image of the sample.

[0023] Relative dimensions of components in drawings may be exaggerated for clarity. Within the following description of drawings, the same or like reference numbers refer to the same or like components or entities, and only the differences with respect to the individual embodiments are described.

[0024] As used herein, unless specifically stated otherwise, the term “or” encompasses all possible combinations, except where infeasible. For example, if it is stated that a component may include A or B, then, unless specifically stated otherwise or infeasible, the component may include A, or B, or A and B. As a second example, if it is stated that a component may include A, B, or C, then, unless specifically stated otherwise or infeasible, the component may include A, or B, or C, or A and B, or A and C, or B and C, or A and B and C.

[0025] FIG. 1 illustrates an exemplary electron beam inspection (EBI) system **100** consistent with embodiments of the present disclosure. EBI system **100** may be used for imaging. As shown in FIG. 1, EBI system **100** includes a main chamber **101**, a load/lock chamber **102**, an electron beam tool **104**, and an equipment front end module (EFEM) **106**. Electron beam tool **104** is located within main chamber **101**. EFEM **106** includes a first loading port **106a** and a second loading port **106b**. EFEM **106** may include additional loading port(s). First loading port **106a** and second loading port **106b** receive wafer front opening unified pods (FOUPs) that contain wafers (e.g., semiconductor wafers or wafers made of other material(s)) or samples to be inspected (wafers and samples may be used interchangeably). A “lot” is a plurality of wafers that may be loaded for processing as a batch.

[0026] One or more robotic arms (not shown) in EFEM **106** may transport the wafers to load/lock chamber **102**. Load/lock chamber **102** is connected to a load/lock vacuum pump system (not shown) which removes gas molecules in load/lock chamber **102** to reach a first pressure below the atmospheric pressure. After reaching the first pressure, one or more robotic arms (not shown) may transport the wafer from load/lock chamber **102** to main chamber **101**. Main chamber **101** is connected to a main chamber vacuum pump system (not shown) which removes gas molecules in main chamber **101** to reach a second pressure below the first pressure. After reaching the second pressure, the wafer is subject to inspection by electron beam tool **104**. Electron beam tool **104** may be a single-beam system or a multi-beam system.

[0027] A controller **109** is electronically connected to electron beam tool **104**. Controller **109** may be a computer configured to execute various controls of EBI system **100**. While controller **109** is shown in FIG. 1 as being outside of the structure that includes main chamber **101**, load/lock chamber **102**, and EFEM **106**, it is appreciated that controller **109** may be a part of the structure.

[0028] In some embodiments, controller **109** may include one or more processors (not shown). A processor may be a generic or specific electronic device capable of manipulating or processing information. For example, the processor may include any combination of any number of a central processing unit (or “CPU”), a graphics processing unit (or “GPU”), an optical processor, a programmable logic controllers, a microcontroller, a microprocessor, a digital signal processor, an intellectual property (IP) core, a Programmable Logic Array (PLA), a Programmable Array Logic (PAL), a Generic Array Logic (GAL), a Complex Programmable Logic Device (CPLD), a Field-Programmable Gate Array (FPGA), a System On Chip (SoC), an Application-Specific Integrated Circuit (ASIC), and any type circuit capable of data processing. The processor may also be a virtual processor that includes one or more processors distributed across multiple machines or devices coupled via a network.

[0029] In some embodiments, controller **109** may further include one or more memories (not shown). A memory may be a generic or specific electronic device capable of storing codes and data accessible by the processor (e.g., via a bus). For example, the memory may include any combination of any number of a random-access memory (RAM), a read-only memory (ROM), an optical disc, a magnetic disk, a hard drive, a solid-state drive, a flash drive, a security digital (SD) card, a memory stick, a compact flash (CF) card, or any type of storage device. The codes may include an operating system (OS) and one or more application programs (or “apps”) for specific tasks. The memory may also be a virtual memory that includes one or more memories distributed across multiple machines or devices coupled via a network.

[0030] Reference is now made to FIG. 2, which is a schematic diagram illustrating an exemplary electron beam tool **104** including a multi-beam inspection tool that is part of the EBI system **100** of FIG. 1, consistent with embodiments of the present disclosure. In some embodiments, electron beam tool **104** may be operated as a single-beam inspection tool that is part of EBI system **100** of FIG. 1. Multi-beam electron beam tool **104** (also referred to herein as apparatus **104**) comprises an electron source **201**, a Coulomb aperture plate (or “gun aperture plate”) **271**, a condenser lens **210**, a source conversion unit **220**, a primary projection system **230**, a motorized stage **209**, and a sample holder **207** supported by motorized stage **209** to hold a sample **208** (e.g., a wafer or a photomask) to be inspected. Multi-beam electron beam tool **104** may further comprise a secondary projection system **250** and an electron detection device **240**. Primary projection system **230** may comprise an objective lens **231**. Electron detection device **240** may comprise a plurality of detection elements **241**, **242**, and **243**. A beam separator **233** and a deflection scanning unit **232** may be positioned inside primary projection system **230**.

[0031] Electron source **201**, Coulomb aperture plate **271**, condenser lens **210**, source conversion unit **220**, beam separator **233**, deflection scanning unit **232**, and primary projection system **230** may be aligned with a primary optical axis **204** of apparatus **104**. Secondary projection system **250** and electron detection device **240** may be aligned with a secondary optical axis **251** of apparatus **104**.

[0032] Electron source **201** may comprise a cathode (not shown) and an extractor or anode (not shown), in which, during operation, electron source **201** is configured to emit primary electrons from the cathode and the primary electrons are extracted or accelerated by the extractor and/or the anode to form a primary electron beam **202** that form a primary beam crossover (virtual or real) **203**. Primary electron beam **202** may be visualized as being emitted from primary beam crossover **203**.

[0033] Source conversion unit **220** may comprise an image-forming element array (not shown), an aberration compensator array (not shown), a beam-limit aperture array (not shown), and a pre-bending micro-deflector array (not shown). In some embodiments, the pre-bending micro-deflector array deflects a plurality of primary beamlets **211**, **212**, **213** of primary electron beam **202** to normally enter the beam-limit aperture array, the image-forming element array, and an aberration compensator array. In some embodiments, apparatus **104** may be operated as a single-beam system such that a single primary beamlet is generated. In some embodiments, condenser lens **210** is designed to focus primary electron beam **202** to become a parallel beam and be normally incident onto source conversion unit **220**. The image-forming element array may comprise a plurality of micro-deflectors or micro-lenses to influence the plurality of primary beamlets **211**, **212**, **213** of primary electron beam **202** and to form a plurality of parallel images (virtual or real) of primary beam crossover **203**, one for each of the primary beamlets **211**, **212**, and **213**. In some embodiments, the aberration compensator array may comprise a field curvature compensator array (not shown) and an astigmatism compensator array (not shown). The field curvature compensator array may comprise a plurality of micro-lenses to compensate field curvature aberrations of the primary beamlets **211**, **212**, and **213**. The astigmatism compensator array may comprise a plurality of micro-stigmators to compensate astigmatism aberrations of the primary beamlets **211**, **212**, and **213**. The beam-limit aperture array may be configured to limit diameters of individual primary

beamlets **211**, **212**, and **213**. FIG. 2 shows three primary beamlets **211**, **212**, and **213** as an example, and it is appreciated that source conversion unit **220** may be configured to form any number of primary beamlets. Controller **109** may be connected to various parts of EBI system **100** of FIG. 1, such as source conversion unit **220**, electron detection device **240**, primary projection system **230**, or motorized stage **209**. In some embodiments, as explained in further details below, controller **109** may perform various image and signal processing functions. Controller **109** may also generate various control signals to govern operations of the charged particle beam inspection system.

[0034] Condenser lens **210** is configured to focus primary electron beam **202**. Condenser lens **210** may further be configured to adjust electric currents of primary beamlets **211**, **212**, and **213** downstream of source conversion unit **220** by varying the focusing power of condenser lens **210**. Alternatively, the electric currents may be changed by altering the radial sizes of beam-limit apertures within the beam-limit aperture array corresponding to the individual primary beamlets. The electric currents may be changed by both altering the radial sizes of beam-limit apertures and the focusing power of condenser lens **210**. Condenser lens **210** may be an adjustable condenser lens that may be configured so that the position of its first principle plane is movable. The adjustable condenser lens may be configured to be magnetic, which may result in off-axis beamlets **212** and **213** illuminating source conversion unit **220** with rotation angles. The rotation angles change with the focusing power or the position of the first principal plane of the adjustable condenser lens. Condenser lens **210** may be an anti-rotation condenser lens that may be configured to keep the rotation angles unchanged while the focusing power of condenser lens **210** is changed. In some embodiments, condenser lens **210** may be an adjustable anti-rotation condenser lens, in which the rotation angles do not change when its focusing power and the position of its first principal plane are varied.

[0035] Objective lens **231** may be configured to focus beamlets **211**, **212**, and **213** onto a sample **208** for inspection and may form, in the current embodiments, three probe spots **221**, **222**, and **223** on the surface of sample **208**. Coulomb aperture plate **271**, in operation, is configured to block off peripheral electrons of primary electron beam **202** to reduce Coulomb effect. The Coulomb effect may enlarge the size of each of probe spots **221**, **222**, and **223** of primary beamlets **211**, **212**, **213**, and therefore deteriorate inspection resolution.

[0036] Beam separator **233** may, for example, be a Wien filter comprising an electrostatic deflector generating an electrostatic dipole field and a magnetic dipole field (not shown in FIG. 2). In operation, beam separator **233** may be configured to exert an electrostatic force by electrostatic dipole field on individual electrons of primary beamlets **211**, **212**, and **213**. The electrostatic force is equal in magnitude but opposite in direction to the magnetic force exerted by magnetic dipole field of beam separator **233** on the individual electrons. Primary beamlets **211**, **212**, and **213** may therefore pass at least substantially straight through beam separator **233** with at least substantially zero deflection angles.

[0037] Deflection scanning unit **232**, in operation, is configured to deflect primary beamlets **211**, **212**, and **213** to scan probe spots **221**, **222**, and **223** across individual scanning areas in a section of the surface of sample **208**. In response to incidence of primary beamlets **211**, **212**, and **213** or probe spots **221**, **222**, and **223** on sample **208**, electrons emerge from sample **208** and generate three secondary electron beams **261**, **262**, and **263**. Each of secondary electron beams **261**, **262**, and **263** typically comprise secondary electrons (having electron energy ≤ 50 eV) and backscattered electrons (having electron energy between 50 eV and the landing energy of primary beamlets **211**, **212**, and **213**). Beam separator **233** is configured to deflect secondary electron beams **261**, **262**, and **263** towards secondary projection system **250**. Secondary projection system **250** subsequently focuses secondary electron beams **261**, **262**, and **263** onto detection elements **241**, **242**, and **243** of electron detection device **240**. Detection elements **241**, **242**, and **243** are arranged to detect corresponding secondary electron beams **261**, **262**, and **263** and generate corresponding signals which are sent to controller **109** or a signal processing system (not shown), e.g., to construct

images of the corresponding scanned areas of sample **208**.

[0038] In some embodiments, detection elements **241**, **242**, and **243** detect corresponding secondary electron beams **261**, **262**, and **263**, respectively, and generate corresponding intensity signal outputs (not shown) to an image processing system (e.g., controller **109**). In some embodiments, each detection element **241**, **242**, and **243** may comprise one or more pixels. The intensity signal output of a detection element may be a sum of signals generated by all the pixels within the detection element.

[0039] In some embodiments, controller **109** may comprise image processing system that includes an image acquirer (not shown), a storage (not shown). The image acquirer may comprise one or more processors. For example, the image acquirer may comprise a computer, server, mainframe host, terminals, personal computer, any kind of mobile computing devices, and the like, or a combination thereof. The image acquirer may be communicatively coupled to electron detection device **240** of apparatus **104** through a medium such as an electrical conductor, optical fiber cable, portable storage media, IR, Bluetooth, internet, wireless network, wireless radio, among others, or a combination thereof. In some embodiments, the image acquirer may receive a signal from electron detection device **240** and may construct an image. The image acquirer may thus acquire images of sample **208**. The image acquirer may also perform various post-processing functions, such as generating contours, superimposing indicators on an acquired image, and the like. The image acquirer may be configured to perform adjustments of brightness and contrast, etc. of acquired images. In some embodiments, the storage may be a storage medium such as a hard disk, flash drive, cloud storage, random access memory (RAM), other types of computer readable memory, and the like. The storage may be coupled with the image acquirer and may be used for saving scanned raw image data as original images, and post-processed images.

[0040] In some embodiments, the image acquirer may acquire one or more images of a sample based on an imaging signal received from electron detection device **240**. An imaging signal may correspond to a scanning operation for conducting charged particle imaging. An acquired image may be a single image comprising a plurality of imaging areas. The single image may be stored in the storage. The single image may be an original image that may be divided into a plurality of regions. Each of the regions may comprise one imaging area containing a feature of sample **208**. The acquired images may comprise multiple images of a single imaging area of sample **208** sampled multiple times over a time sequence. The multiple images may be stored in the storage. In some embodiments, controller **109** may be configured to perform image processing steps with the multiple images of the same location of sample **208**.

[0041] In some embodiments, controller **109** may include measurement circuitries (e.g., analog-to-digital converters) to obtain a distribution of the detected secondary electrons. The electron distribution data collected during a detection time window, in combination with corresponding scan path data of each of primary beamlets **211**, **212**, and **213** incident on the wafer surface, can be used to reconstruct images of the wafer structures under inspection. The reconstructed images can be used to reveal various features of the internal or external structures of sample **208**, and thereby can be used to reveal any defects that may exist in the wafer.

[0042] In some embodiments, controller **109** may control motorized stage **209** to move sample **208** during inspection of sample **208**. In some embodiments, controller **109** may enable motorized stage **209** to move sample **208** in a direction continuously at a constant speed. In other embodiments, controller **109** may enable motorized stage **209** to change the speed of the movement of sample **208** overtime depending on the steps of scanning process.

[0043] Although FIG. **2** shows that apparatus **104** uses three primary electron beams, it is appreciated that apparatus **104** may use two or more number of primary electron beams. The present disclosure does not limit the number of primary electron beams used in apparatus **104**. In some embodiments, apparatus **104** may be a SEM used for lithography. In some embodiments, electron beam tool **104** may be a single-beam system or a multi-beam system.

[0044] Compared with a single charged-particle beam imaging system (“single-beam system”), a multiple charged-particle beam imaging system (“multi-beam system”) may be designed to optimize throughput for different scan modes. Embodiments of this disclosure provide a multi-beam system with the capability of optimizing throughput for different scan modes by using beam arrays with different geometries, adapting to different throughputs and resolution requirements. [0045] FIG. 3 is a schematic diagram of a system for improving image quality, consistent with embodiments of the present disclosure. System **300** may include an inspection system **310** and an image generation component **320**. Inspection system **310** and image generation component **320** may be electrically coupled (directly or indirectly) to each other, either physically (e.g., by a cable) or remotely. Inspection system **310** may be the system described with respect to FIGS. 1 and 2, used to acquire images of a wafer (see, e.g., sample **208** of FIG. 2). In some embodiments, components of system **300** may be implemented as one or more servers (e.g., where each server includes its own processor). In some embodiments, components of system **300** may be implemented as software that may pull data from one or more databases of system **300**. In some embodiments, system **300** may include one server or a plurality of servers. In some embodiments, system **300** may include one or more modules that are implemented by a controller (e.g., controller **109** of FIG. 1, controller **109** of FIG. 2).

[0046] Inspection system **310** may obtain a plurality of images (e.g., images **452** of FIG. 4) of an area (e.g., areas **411** or **412** of FIG. 4) of a sample (e.g., sample **208** of FIG. 2 or sample **400** of FIG. 4). Each obtained image of the plurality of images may have a different focus-related value (e.g., defocus value) due to an adjustment of an inspection setting during image acquisition. For example, the adjusted inspection setting may be a strength of the objective lens (e.g., using different voltage values or different current values). In some embodiments, the objective lens (e.g., objective lens **231** of FIG. 2) may be adjusted before each image of an area of a FOV (e.g., FOV **410** of FIG. 4) is obtained such that each image has a different defocus value. In some embodiments, the adjusted inspection setting may be a range of strength of the objective lens (e.g., a range of voltage values). Each image of the plurality of images may be obtained from the same area of a sample in a FOV. Inspection system **310** may transmit data including the plurality of images of the area of the sample to image generation component **320**.

[0047] Image generation component **320** may include one or more processors (e.g., represented as processor **322**, which can have one or more corresponding accelerators) and a storage **324**. Image generation component **320** may also include a communication interface **326** to receive from and send data to inspection system **310**. In some embodiments, processor **322** may be configured to use a phase diversity analysis to determine a defocus value associated with each image of the plurality of images received from inspection system **310** and a maximum likelihood estimate of the plurality of images.

[0048] A phase diversity analysis involves using multiple images of the same area, where a phase diversity (e.g., an aberration, a defocused electron beam, etc.) may be introduced to the area before each image of the multiple images is obtained. For example, a phase diversity may be introduced to the area before each image is obtained by adjusting the strength of the objective lens before obtaining each image. By controlling the phase diversity that is introduced to the area, processor **322** may maximize an objective function and solve for the unknown variables (e.g., defocus value of an image).

[0049] A maximum likelihood estimate of the plurality of images may be obtained from the defocused images and their associated defocus values using phase diversity analysis. In some embodiments, the maximum likelihood estimate may be a final, focus-adjusted (e.g., focus-corrected) image of an area within the FOV. The maximum likelihood estimate may be obtained regardless of whether the defocus values correspond to the true unknown defocus value since the defocus distance between the images may be known. The phase diversity analysis allows for simultaneous determination of the maximum likelihood estimate as well as the true unknown

defocus value. Image generation component **320** may advantageously use a phase diversity analysis with a non-iterative approach to maximize the objective function since it solves for a single unknown variable (e.g., defocus value of an image), thereby increasing throughput of inspection. [0050] In some embodiments, other types of phase diversity may be used in the phase diversity analysis. For example, phase diversity types that may be used in addition to, or in place of, a defocus value include aperture shifts, detuning via a stigmator, using beam deflectors to probe different parts of the generalized pupil function characterizing the electro optical column, etc.

[0051] In some embodiments, higher order aberrations may be used in a phase diversity analysis (e.g., astigmatism). In some cases, higher order aberrations or additional types of phase diversities may not be preferred since they may involve solving for more unknown variables, thereby increasing computational load, increasing throughput, and reducing precision of generating a corrected image. In some embodiments, these constraints may be reduced by increasing the number of images obtained from an area in the FOV.

[0052] The described embodiments advantageously generate an image of a sample (e.g., for inspection) by obtaining images within a FOV without obtaining images outside of the FOV (e.g., in area **420** of FIG. **4**), thereby increasing throughput.

[0053] In some embodiments, the above-described methods may be performed for a plurality of areas (e.g., areas **411** and **412** of FIG. **4**) within a FOV. For example, a first phase diversity analysis may be performed for a first set of images associated with a first area (e.g., area **411** of FIG. **4**) of a sample in the FOV. A second phase diversity analysis may be performed for a second set of images associated with a second area (e.g., area **412** of FIG. **4**) of the sample in the FOV, where the first area may be different (e.g., a separate area) from the second area. In some embodiments, an adjustment of an inspection setting (e.g., using different voltage values or different current values, a range of voltage values, etc.) for the first area may be the same adjustment of an inspection setting for the second area (e.g., the voltage values used for obtaining images from the first area are the same as the voltage values used for obtaining images from the second area). It should be understood that using the same adjustment of an inspection setting for different areas does not necessarily mean that the obtained images for both areas will have the same associated focus-related values. In some embodiments, an adjustment of an inspection setting for the first area may be different from the adjustment of an inspection setting of the second area (e.g., one or more of the voltage values used for obtaining images from the first area is different from one or more of the voltage values used for obtaining images from the second area).

[0054] In some embodiments, performing phase diversity analyses to generate images for different areas of a sample may increase the accuracy of the calculations by performing local calculations instead of applying the same inspection settings (e.g., objective lens strength) across an entire FOV, thereby increasing the quality of the generated images. For example, to generate higher quality images of a sample in a FOV, the voltage setting used on the objective lens may vary depending on the area in the FOV that is being inspected. That is, the above-described methods may be performed for one or more areas within a FOV so that a focus-corrected image that is local to (e.g., specific to) the area may be generated. This method may be advantageous over a method that uses the same inspection settings (e.g., objective lens voltage) over the entire FOV to generate images. Additionally, the determination of the focus value for each area may provide valuable diagnostic data to improve the control or design of the imaging system.

[0055] This method may advantageously increase the image quality of a generated image (e.g., image **454** of FIG. **4**) since focus-corrected images of an area may be generated by obtaining images of the same area, rather than depending on focus measurements performed outside of the FOV.

[0056] By generating a focus-corrected image of a sample area within the FOV using images obtained from within the FOV, this method may advantageously increase the image quality of the generated image by counteracting focus deviations that may occur over a sample (e.g., the field

curvature may vary across a FOV). For example, the focus of an electron beam may change as the electron beam moves further away from the optical axis (e.g., the center of the FOV).

[0057] For example, a first area (e.g., area **411** of FIG. **4**) of a sample may be located near the center of the FOV and a second area (e.g., area **412** of FIG. **4**) of the sample may be located away from the center of the FOV. A first phase diversity analysis may be performed to generate an inspection image of the first area using images obtained from the first area. A second phase diversity analysis may be performed to generate an inspection image of the second area using images obtained from the second area. The disclosed embodiments include generation of images that overcome the constraints of focus deviations over a FOV of a sample.

[0058] In some embodiments, one or more obtained images may include noise that reduces the accuracy of the calculated defocus values or the accuracy of the maximum likelihood estimate, thereby reducing the quality of the generated one or more images. In some embodiments, due to the low dosage electron beam(s) (e.g., low number of nominal incident primary electrons per pixel) used to obtain an image of an area, the noise in the obtained images may be modeled by a Poisson distribution. The obtained images may be denoised, for example, using a method such as methods used in Zhang et al., “Modified phase diversity technique to eliminate Poisson noise for reconstructing high-resolution images”, Proc. SPIE 10838, 2019 Jan. 16. The residual noise in the images may be characterized by a Gaussian distribution.

[0059] In some embodiments, processor **322** may use a phase diversity analysis to determine a defocus value associated with each denoised image of the plurality of denoised images and a maximum likelihood estimate of the plurality of denoised images. The phase diversity analysis may be performed consistent with the disclosed embodiments.

[0060] By denoising the obtained images modeled after a Poisson distribution and characterizing the residual noise in the images following a Gaussian distribution, this method may advantageously reduce the computational load required during a phase diversity analysis, thereby resulting in higher quality images and increasing throughput. The phase diversity analysis may be performed on the denoised images using a method such as methods used in Zhang et al., “Modified phase diversity technique to eliminate Poisson noise for reconstructing high-resolution images”, Proc. SPIE 10838, 2019 Jan. 16. This method may be applied without denoising in the low dosage case

[0061] Reference is now made to FIG. **4**, a schematic diagram illustrating an exemplary sample **400** and an image generation schematic **450**, consistent with embodiments of the present disclosure.

[0062] As described above, an inspection system (e.g., inspection system **310** of FIG. **3**) may obtain a plurality of images **452** of an area **411** of a sample (e.g., sample **208** of FIG. **2**). Each obtained image of plurality of images **452** may have a different focus-related value (e.g., defocus value) due to an adjustment of an inspection setting during image acquisition. For example, the adjusted inspection setting may be a strength of the objective lens (e.g., using different voltage values). In some embodiments, the objective lens may be adjusted before each image of area **411** of a FOV **410** is obtained such that each image has a different defocus value. In some embodiments, the adjusted inspection setting may be a range of strength of the objective lens (e.g., a range of voltage values). Each image of plurality of images **452** may be obtained from the same area **411** of a sample in FOV **420**.

[0063] As described above, an image generation component (e.g., image generation component **320** of FIG. **3**) may include a processor (e.g., process **322** of FIG. **3**) configured to use a phase diversity analysis to determine a defocus value associated with each image of plurality of images **452** and a maximum likelihood estimate of plurality of images **452**.

[0064] A maximum likelihood estimate of plurality of images **452** may be obtained from the defocused images and their associated defocus values using phase diversity analysis. In some embodiments, the maximum likelihood estimate may be a final, focus-adjusted (e.g., focus-corrected) image **454** of area **411** within FOV **410**. The maximum likelihood estimate may be

obtained regardless of whether the defocus values correspond to the true unknown defocus value since the defocus distance between the images may be known. The phase diversity analysis allows for simultaneous determination of the maximum likelihood estimate as well as the true unknown defocus value. The image generation component may advantageously use a phase diversity analysis with a non-iterative approach to maximize the objective function since it solves for a single unknown variable (e.g., defocus value of an image), thereby increasing throughput of inspection. [0065] The described embodiments advantageously generate image **454** of sample **400** (e.g., for inspection) by obtaining images **452** within FOV **410** without obtaining images from an area **420** outside of FOV **410**, thereby increasing throughput.

[0066] In some embodiments, the above-described methods may be performed for a plurality of areas **411** and **412** within FOV **410**. For example, a first phase diversity analysis may be performed for a first set of images associated with first area **411** of sample **400** in FOV **420**. A second phase diversity analysis may be performed for a second set of images associated with second area **412** of sample **400** in FOV **410**, where first area **411** may be different (e.g., a separate area) from second area **412**.

[0067] In some embodiments, performing phase diversity analyses to generate images for different areas of a sample may increase the accuracy of the calculations by performing local calculations instead of applying the same inspection settings (e.g., objective lens strength) across an entire FOV **410**, thereby increasing the quality of the generated images. For example, to generate higher quality images of sample **400** in FOV **410**, the voltage setting used on the objective lens may vary depending on the area in FOV **410** that is being inspected. That is, the above-described methods may be performed for one or more areas within FOV **410** so that a focus-corrected image that is local to (e.g., specific to) the area may be generated. This method may be advantageous over a method that uses the same inspection settings (e.g., objective lens voltage) over the entire FOV **410** to generate images. For example, this method may use a first inspection setting to obtain images and generate a focus-corrected image of area **411** and a second inspection setting to obtain images and generate a focus-corrected image of area **412**, as opposed to using the same inspection settings for areas **411** and **412** without performing separate analyses on both areas. Additionally, the determination of the focus value for each area may provide valuable diagnostic data to improve the control or design of the imaging system.

[0068] This method may advantageously increase the image quality of a generated image (e.g., image **454** of FIG. **4**) since focus-corrected images of an area may be generated by obtaining images of the same area, rather than depending on focus measurements performed in area **420** outside of FOV **410**.

[0069] By generating a focus-corrected image of a sample area within FOV **410** using images obtained from within FOV **410**, this method may advantageously increase the image quality of the generated image by counteracting focus deviations that may occur over a sample (e.g., the field curvature may vary across a FOV). For example, the focus of an electron beam over area **411** may be different from the focus of an electron beam over area **412**.

[0070] For example, area **411** of sample **400** may be located near the center of FOV **410** and area **412** of sample **400** may be located away from the center of FOV **410**. A first phase diversity analysis may be performed to generate an inspection image of area **411** using images obtained from area **411**. A second phase diversity analysis may be performed to generate an inspection image of area **412** using images obtained from area **412**. The disclosed embodiments include generation of images that overcome the constraints of focus deviations over a FOV of a sample. In some embodiments, a single beam may be used to obtain images from one or more areas of the sample within the FOV. In some embodiments, two or more areas of the sample within the FOV may be scanned by different beams.

[0071] In some embodiments, one or more obtained images **452** may include noise that reduces the accuracy of the calculated defocus values or the accuracy of the maximum likelihood estimate,

thereby reducing the quality of generated one or more images **454**. In some embodiments, due to the low dosage electron beam(s) (e.g., low number of nominal incident primary electrons per pixel) used to obtain an image of an area, the noise in the obtained images may be modeled by a Poisson distribution. The obtained images **452** may be denoised, for example, using a method such as methods used in Zhang et al., “Modified phase diversity technique to eliminate Poisson noise for reconstructing high-resolution images”, Proc. SPIE 10838, 2019 Jan. 16. The residual noise in the images may be characterized by a Gaussian distribution.

[0072] In some embodiments, a processor may use a phase diversity analysis to determine a defocus value associated with each denoised image of the plurality of denoised images **452** and a maximum likelihood estimate of the plurality of denoised images **452**. The phase diversity analysis may be performed consistent with the disclosed embodiments.

[0073] By denoising the obtained images **452** modeled after a Poisson distribution and characterizing the residual noise in images **452** following a Gaussian distribution, this method may advantageously reduce the computational load required during a phase diversity analysis, thereby resulting in higher quality images **454** and increasing throughput. The phase diversity analysis may be performed on the denoised images **452** using a method such as methods used in Zhang et al., “Modified phase diversity technique to eliminate Poisson noise for reconstructing high-resolution images”, Proc. SPIE 10838, 2019 Jan. 16. This method may be applied without denoising in the low dosage case.

[0074] Reference is now made to FIG. 5, a flowchart illustrating an exemplary process **500** of improving image quality, consistent with embodiments of the present disclosure. The steps of method **500** can be performed by a system (e.g., system **300** of FIG. 3) executing on or otherwise using the features of a computing device, e.g., controller **109** of FIG. 1 for purposes of illustration. It is appreciated that the illustrated method **500** can be altered to modify the order of steps and to include additional steps.

[0075] At step **501**, the system (e.g., using inspection system **310** of FIG. 3) may obtain a plurality of images e.g., images **452** of FIG. 4) of an area (e.g., areas **411** or **412** of FIG. 4) of a sample (e.g., sample **208** of FIG. 2 or sample **400** of FIG. 4). Each obtained image of the plurality of images may have a different focus-related value (e.g., defocus value) due to an adjustment of an inspection setting during image acquisition. For example, the adjusted inspection setting may be a strength of the objective lens (e.g., using different voltage values). In some embodiments, the objective lens (e.g., objective lens **231** of FIG. 2) may be adjusted before each image of an area of a FOV (e.g., FOV **410** of FIG. 4) is obtained such that each image has a different defocus value. In some embodiments, the adjusted inspection setting may be a range of strength of the objective lens (e.g., a range of voltage values). Each image of the plurality of images may be obtained from the same area of a sample in a FOV.

[0076] At step **503**, a processor (e.g., processor **322** of FIG. 3) may be configured to use a phase diversity analysis to determine a defocus value associated with each image of the plurality of images received from the inspection system and a maximum likelihood estimate of the plurality of images.

[0077] A phase diversity analysis involves using multiple images of the same area, where a phase diversity (e.g., an aberration, a defocused electron beam, etc.) may be introduced to the area before each image of the multiple images is obtained. For example, a phase diversity may be introduced to the area before each image is obtained by adjusting the strength of the objective lens before obtaining each image. By controlling the phase diversity that is introduced to the area, processor **322** may maximize an objective function and solve for the unknown variables (e.g., defocus value of an image). A maximum likelihood estimate of the plurality of images may be obtained from the defocus values associated with each image using the phase diversity analysis.

[0078] In some embodiments, other types of phase diversity may be used in the phase diversity analysis. For example, phase diversity types that may be used in addition to, or in place of, a

defocus value include aperture shifts, detuning via a stigmator, using beam deflectors to probe different parts of the generalized pupil function characterizing the electro optical column, etc. [0079] In some embodiments, higher order aberrations may be used in a phase diversity analysis (e.g., astigmatism). In some cases, higher order aberrations or additional types of phase diversities may not be preferred since they may involve solving for more unknown variables, thereby increasing computational load, increasing throughput, and reducing precision of generating a corrected image. In some embodiments, these constraints may be reduced by increasing the number of images obtained from an area in the FOV.

[0080] At step **505**, a processor may generate a focus-adjusted (e.g., focus-corrected) image (e.g., image **454** of FIG. **4**) of the area based on the determined plurality of defocus values and the determined maximum likelihood estimate. In some embodiments, the maximum likelihood estimate may be a final, focus-corrected image (e.g., image **454** of FIG. **4**) of an area within the FOV. The maximum likelihood estimate may be obtained regardless of whether the defocus values correspond to the true unknown defocus value since the defocus distance between the images may be known. The phase diversity analysis allows for simultaneous determination of the maximum likelihood estimate as well as the true unknown defocus value. The processor advantageously uses a phase diversity analysis with a non-iterative approach to maximize the objective function since it solves for a single unknown variable (e.g., defocus value of an image), thereby increasing throughput of inspection.

[0081] The described embodiments advantageously generate an image of a sample (e.g., for inspection) by obtaining images with a FOV without obtaining images outside of the FOV (e.g., in an area **420** outside of the FOV of FIG. **4**), thereby increasing throughput.

[0082] In some embodiments, the above-described methods may be performed for a plurality of areas (e.g., areas **411** and **412** of FIG. **4**) within a FOV. For example, a first phase diversity analysis may be performed for a first set of images associated with a first area (e.g., area **411** of FIG. **4**) of a sample in the FOV. A second phase diversity analysis may be performed for a second set of images associated with a second area (e.g., area **412** of FIG. **4**) of the sample in the FOV, where the first area may be different (e.g., a separate area) from the second area.

[0083] In some embodiments, performing phase diversity analyses to generate images for different areas of a sample may increase the accuracy of the calculations by performing local calculations instead of applying the same inspection settings (e.g., objective lens strength) across an entire FOV, thereby increasing the quality of the generated images. For example, to generate higher quality images of a sample in a FOV, the voltage setting used on the objective lens may vary depending on the area in the FOV that is being inspected. That is, the above-described methods may be performed for one or more areas within a FOV so that a focus-corrected image that is local to (e.g., specific to) the area may be generated. This method may be advantageous over a method that uses the same inspection settings (e.g., objective lens voltage) over the entire FOV to generate images. Additionally, the determination of the focus value for each area may provide valuable diagnostic data to improve the control or design of the imaging system.

[0084] This method may advantageously increase the image quality of a generated image since focus-corrected images of an area may be generated by obtaining images of the same area, rather than depending on focus measurements performed outside of the FOV.

[0085] By generating a focus-corrected image of a sample area within the FOV using images obtained from within the FOV, this method may advantageously increase the image quality of the generated image by counteracting focus deviations that may occur over a sample (e.g., the field curvature may vary across a FOV). For example, the focus of an electron beam may change as the electron beam moves further away from the optical axis (e.g., the center of the FOV).

[0086] For example, a first area (e.g., area **411** of FIG. **4**) of a sample may be located near the center of the FOV and a second area (e.g., area **412** of FIG. **4**) of the sample may be located away from the center of the FOV. A first phase diversity analysis may be performed to generate an

inspection image of the first area using images obtained from the first area. A second phase diversity analysis may be performed to generate an inspection image of the second area using images obtained from the second area. The disclosed embodiments include generation of images that overcome the constraints of focus deviations over a FOV of a sample.

[0087] In some embodiments, one or more obtained images may include noise that reduces the accuracy of the calculated defocus values or the accuracy of the maximum likelihood estimate, thereby reducing the quality of the generated one or more images. In some embodiments, due to the low dosage electron beam(s) (e.g., low number of nominal incident primary electrons per pixel) used to obtain an image of an area, the noise in the obtained images may be modeled by a Poisson distribution. The obtained images may be denoised, for example, using a method such as methods used in Zhang et al., “Modified phase diversity technique to eliminate Poisson noise for reconstructing high-resolution images”, Proc. SPIE 10838, 2019 Jan. 16. The residual noise in the images may be characterized by a Gaussian distribution.

[0088] In some embodiments, a processor may use a phase diversity analysis to determine a defocus value associated with each denoised image of the plurality of denoised images and a maximum likelihood estimate of the plurality of denoised images. The phase diversity analysis may be performed consistent with the disclosed embodiments.

[0089] By denoising the obtained images modeled after a Poisson distribution and characterizing the residual noise in the images following a Gaussian distribution, this method may advantageously reduce the computational load required during a phase diversity analysis, thereby resulting in higher quality images and increasing throughput. The phase diversity analysis may be performed on the denoised images using a method such as methods used in Zhang et al., “Modified phase diversity technique to eliminate Poisson noise for reconstructing high-resolution images”, Proc. SPIE 10838, 2019 Jan. 16. This method may be applied without denoising in the low dosage case.

[0090] A non-transitory computer readable medium may be provided that stores instructions for a processor of a controller (e.g., controller **109** of FIG. **1**) for controlling the electron beam tool, controlling the voltages applies to the objective lens (e.g., objective lens **231** of FIG. **2**), or controlling processors (e.g., processor **322** of FIG. **3**) of other systems and servers, consistent with embodiments in the present disclosure. These instructions may allow the one or more processors to carry out image processing, data processing, beamlet scanning, database management, graphical display, operations of a charged particle beam apparatus, or another imaging device, or the like. In some embodiments, the non-transitory computer readable medium may be provided that stores instructions for a processor to perform the steps of process **500**. Common forms of non-transitory media include, for example, a floppy disk, a flexible disk, hard disk, solid state drive, magnetic tape, or any other magnetic data storage medium, a Compact Disc Read Only Memory (CD-ROM), any other optical data storage medium, any physical medium with patterns of holes, a Random Access Memory (RAM), a Programmable Read Only Memory (PROM), and Erasable Programmable Read Only Memory (EPROM), a FLASH-EPROM or any other flash memory, Non-Volatile Random Access Memory (NVRAM), a cache, a register, any other memory chip or cartridge, and networked versions of the same.

[0091] The embodiments may further be described using the following clauses: [0092] 1. A method of improving image quality, the method comprising: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined MLE. [0093] 2. The method of clause 1, wherein each focus-related value of the plurality of focus-related values comprises a defocus value. [0094] 3. The method of any one of clauses 1-2, wherein each image of the plurality of images has a different associated focus-related value. [0095] 4. The method of any one of clauses 1-3, wherein the plurality of focus-

related values comprise a range of focus-related values. [0096] 5. The method of any one of clauses 1-4, wherein the range of focus-related values corresponds to a range of voltages associated with an objective lens. [0097] 6. The method of any one of clauses 1-5, wherein the area is within a field of view of the sample. [0098] 7. The method of any one of clauses 1-6, wherein the MLE is determined based on the determined plurality of focus-related values. [0099] 8. The method of any one of clauses 1-7, wherein: [0100] the area comprises a first area of the sample and a second area of the sample; [0101] the plurality of images comprises a first set of images of the first area of the sample and a second set of images of the second area of the sample; the phase diversity analysis comprises a first phase diversity analysis corresponding to the first set of images and a second phase diversity analysis corresponding to the second set of images; the plurality of focus-related values comprises a first set of focus-related values corresponding to the first phase diversity analysis and a second set of focus-related values corresponding to the second phase diversity analysis; [0102] the MLE comprises a first MLE of the first set of images and a second MLE of the second set of images; [0103] and the focus-corrected image comprises a first focus-corrected image of the first area and a second focus-corrected image of the second area. [0104] 9. The method of clause 8, further comprising: determining, via the first phase diversity analysis: the first set of focus-related values, wherein each focus-related value of the first set of focus-related values is associated with each image of the first set of images; the first MLE of the first set of images; [0105] and generating the first focus-corrected image of the first area based on the determined first plurality of focus-related values and the determined first MLE. [0106] 10. The method of any one of clauses 8-9, further comprising: determining, via the second phase diversity analysis: the second set of focus-related values, wherein each focus-related value of the second set of focus-related values is associated with each image of the second set of images; the second MLE of the second set of images; and generating the second focus-corrected image of the second area based on the determined second plurality of focus-related values and the determined second MLE. [0107] 11. The method of any one of clauses 1-10, further comprising: denoising noise in the obtained plurality of images, wherein the noise is modeled as a Poisson distribution; and modeling the denoised noise in the obtained plurality of images on a Gaussian distribution. [0108] 12. The method of clause 11, wherein each focus-related value of the plurality of focus-related values is associated with each denoised image of the plurality of images modeled on the Gaussian distribution. [0109] 13. A method of improving image quality, the method comprising: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-adjusted image of the area based on the determined plurality of focus-related values and the determined MLE. [0110] 14. A method of improving image quality, the method comprising: obtaining a plurality of images of an area in a field of view of a sample, wherein a first image of the plurality of images has a first focus-related value and a second image of the plurality of images has a second focus-related value that is different from the first focus-related value; and generating a focus-adjusted image of the area in the field of view using the first and second images. [0111] 15. The method of clause 14, wherein generating the focus-adjusted image of the area in the field of view comprises: performing a phase diversity analysis of the first and second images; and determining a maximum likelihood estimate of the first and second images. [0112] 16. The method of any one of clauses 14-15, wherein the first focus-related value corresponds to a first voltage associated with an objective lens and the second focus-related value corresponds to a second voltage associated with the objective lens. [0113] 17. The method of any one of clauses 14-16, further comprising: [0114] denoising noise in the first image and the second image, wherein the noise is modeled as a Poisson distribution; and modeling the denoised noise in the first image and the second image on a Gaussian distribution. [0115] 18. The method of clause 17, wherein the first focus-related value is

associated with the first denoised image and the second focus-related value is associated with the second denoised image. [0116] 19. A system for improving image quality, the system comprising: a controller including circuitry configured to cause the system to perform: [0117] obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: [0118] a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined MLE. [0119] 20. The system of clause 19, wherein each focus-related value of the plurality of focus-related values comprises a defocus value. [0120] 21. The system of any one of clauses 19-20, wherein each image of the plurality of images has a different associated focus-related value. [0121] 22. The system of any one of clauses 19-21, wherein the plurality of focus-related values comprise a range of focus-related values. [0122] 23. The system of any one of clauses 19-22, wherein the range of focus-related values corresponds to a range of voltages associated with an objective lens. [0123] 24. The system of any one of clauses 19-23, wherein the area is within a field of view of the sample. [0124] 25. The system of any one of clauses 19-24, wherein the MLE is determined based on the determined plurality of focus-related values. [0125] 26. The system of any one of clauses 19-25, wherein: the area comprises a first area of the sample and a second area of the sample; the plurality of images comprises a first set of images of the first area of the sample and a second set of images of the second area of the sample; the phase diversity analysis comprises a first phase diversity analysis corresponding to the first set of images and a second phase diversity analysis corresponding to the second set of images; the plurality of focus-related values comprises a first set of focus-related values corresponding to the first phase diversity analysis and a second set of focus-related values corresponding to the second phase diversity analysis; the MLE comprises a first MLE of the first set of images and a second MLE of the second set of images; and the focus-corrected image comprises a first focus-corrected image of the first area and a second focus-corrected image of the second area. [0126] 27. The system of clause 26, wherein the circuitry is configured to cause the system to further perform: determining, via the first phase diversity analysis: the first set of focus-related values, wherein each focus-related value of the first set of focus-related values is associated with each image of the first set of images; the first MLE of the first set of images; and generating the first focus-corrected image of the first area based on the determined first plurality of focus-related values and the determined first MLE. [0127] 28. The system of any one of clauses 26-27, wherein the circuitry is configured to cause the system to further perform: determining, via the second phase diversity analysis: the second set of focus-related values, wherein each focus-related value of the second set of focus-related values is associated with each image of the second set of images; the second MLE of the second set of images; and generating the second focus-corrected image of the second area based on the determined second plurality of focus-related values and the determined second MLE. [0128] 29. The system of any one of clauses 19-28, wherein the circuitry is configured to cause the system to further perform: denoising noise in the obtained plurality of images, wherein the noise is modeled as a Poisson distribution; and modeling the denoised noise in the obtained plurality of images on a Gaussian distribution. [0129] 30. The system of clause 29, wherein each focus-related value of the plurality of focus-related values is associated with each denoised image of the plurality of images modeled on the Gaussian distribution. [0130] 31. A system for improving image quality, the system comprising: a controller including circuitry configured to cause the system to perform: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-adjusted image of the area based on the determined plurality of focus-related values and the determined MLE. [0131] 32. A system for improving image quality, the system comprising: a

controller including circuitry configured to cause the system to perform: obtaining a plurality of images of an area in a field of view of a sample, wherein a first image of the plurality of images has a first focus-related value and a second image of the plurality of images has a second focus-related value that is different from the first focus-related value; and generating a focus-adjusted image of the area in the field of view using the first and second images. [0132] 33. The system of clause 32, wherein generating the focus-adjusted image of the area in the field of view comprises: performing a phase diversity analysis of the first and second images; and determining a maximum likelihood estimate of the first and second images. [0133] 34. The system of any one of clauses 32-33, wherein the first focus-related value corresponds to a first voltage associated with an objective lens and the second focus-related value corresponds to a second voltage associated with the objective lens. [0134] 35. The system of any one of clauses 32-34, wherein the circuitry is configured to cause the system to further perform: denoising noise in the first image and the second image, wherein the noise is modeled as a Poisson distribution; and modeling the denoised noise in the first image and the second image on a Gaussian distribution. [0135] 36. The system of clause 35, wherein the first focus-related value is associated with the first denoised image and the second focus-related value is associated with the second denoised image. [0136] 37. A non-transitory computer readable medium that stores a set of instructions that is executable by at least one processor of a computing device to cause the computing device to perform a method for improving image quality, the method comprising: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and [0137] generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined MLE. [0138] 38. The non-transitory computer readable medium of clause 37, wherein each focus-related value of the plurality of focus-related values comprises a defocus value. [0139] 39. The non-transitory computer readable medium of any one of clauses 37-38, wherein each image of the plurality of images has a different associated focus-related value. [0140] 40. The non-transitory computer readable medium of any one of clauses 37-39, wherein the plurality of focus-related values comprise a range of focus-related values. [0141] 41. The non-transitory computer readable medium of any one of clauses 37-40, wherein the range of focus-related values corresponds to a range of voltages associated with an objective lens. [0142] 42. The non-transitory computer readable medium of any one of clauses 37-41, wherein the area is within a field of view of the sample. [0143] 43. The non-transitory computer readable medium of any one of clauses 37-42, wherein the MLE is determined based on the determined plurality of focus-related values. [0144] 44. The non-transitory computer readable medium of any one of clauses 37-43, wherein: [0145] the area comprises a first area of the sample and a second area of the sample; [0146] the plurality of images comprises a first set of images of the first area of the sample and a second set of images of the second area of the sample; the phase diversity analysis comprises a first phase diversity analysis corresponding to the first set of images and a second phase diversity analysis corresponding to the second set of images; the plurality of focus-related values comprises a first set of focus-related values corresponding to the first phase diversity analysis and a second set of focus-related values corresponding to the second phase diversity analysis; the MLE comprises a first MLE of the first set of images and a second MLE of the second set of images; and the focus-corrected image comprises a first focus-corrected image of the first area and a second focus-corrected image of the second area. [0147] 45. The non-transitory computer readable medium of clause 44, wherein the set of instructions that is executable by at least one processor of a computing device to cause the computing device to further perform: determining, via the first phase diversity analysis: the first set of focus-related values, wherein each focus-related value of the first set of focus-related values is associated with each image of the first set of images; the first MLE of the first set of images; and [0148] generating the first focus-corrected image of the first area based on

the determined first plurality of focus-related values and the determined first MLE. [0149] 46. The non-transitory computer readable medium of any one of clauses 44-45, wherein the set of instructions that is executable by at least one processor of a computing device to cause the computing device to further perform: determining, via the second phase diversity analysis: the second set of focus-related values, wherein each focus-related value of the second set of focus-related values is associated with each image of the second set of images; the second MLE of the second set of images; and generating the second focus-corrected image of the second area based on the determined second plurality of focus-related values and the determined second MLE. [0150] 47. The non-transitory computer readable medium of any one of clauses 37-46, wherein the set of instructions that is executable by at least one processor of a computing device to cause the computing device to further perform: denoising noise in the obtained plurality of images, wherein the noise is modeled as a Poisson distribution; and modeling the denoised noise in the obtained plurality of images on a Gaussian distribution. [0151] 48. The non-transitory computer readable medium of clause 47, wherein each focus-related value of the plurality of focus-related values is associated with each denoised image of the plurality of images modeled on the Gaussian distribution. [0152] 49. A non-transitory computer readable medium that stores a set of instructions that is executable by at least one processor of a computing device to cause the computing device to perform a method for improving image quality, the method comprising: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-adjusted image of the area based on the determined plurality of focus-related values and the determined MLE. [0153] 50. A non-transitory computer readable medium that stores a set of instructions that is executable by at least one processor of a computing device to cause the computing device to perform a method for improving image quality, the method comprising: obtaining a plurality of images of an area in a field of view of a sample, wherein a first image of the plurality of images has a first focus-related value and a second image of the plurality of images has a second focus-related value that is different from the first focus-related value; and generating a focus-adjusted image of the area in the field of view using the first and second images. [0154] 51. The non-transitory computer readable medium of clause 50, wherein generating the focus-adjusted image of the area in the field of view comprises: performing a phase diversity analysis of the first and second images; and determining a maximum likelihood estimate of the first and second images. [0155] 52. The non-transitory computer readable medium of any one of clauses 50-51, wherein the first focus-related value corresponds to a first voltage associated with an objective lens and the second focus-related value corresponds to a second voltage associated with the objective lens. [0156] 53. The non-transitory computer readable medium of any one of clauses 50-52, wherein the set of instructions that is executable by at least one processor of a computing device to cause the computing device to further perform: denoising noise in the first image and the second image, wherein the noise is modeled as a Poisson distribution; and modeling the denoised noise in the first image and the second image on a Gaussian distribution. [0157] 54. The non-transitory computer readable medium of clause 53, wherein the first focus-related value is associated with the first denoised image and the second focus-related value is associated with the second denoised image.

[0158] It will be appreciated that the embodiments of the present disclosure are not limited to the exact construction that has been described above and illustrated in the accompanying drawings, and that various modifications and changes may be made without departing from the scope thereof.

Claims

- 1.** A system for improving image quality, the system comprising: a controller including circuitry configured to cause the system to perform: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined MLE.
- 2.** The system of claim 1, wherein each focus-related value of the plurality of focus-related values comprises a defocus value.
- 3.** The system of claim 1, wherein each image of the plurality of images has a different associated focus-related value.
- 4.** The system of claim 1, wherein the plurality of focus-related values comprise a range of focus-related values.
- 5.** The system of claim 1, wherein a range of the plurality of focus-related values corresponds to a range of voltages associated with an objective lens.
- 6.** The system of claim 1, wherein the area is within a field of view of the sample.
- 7.** The system of claim 1, wherein the MLE is determined based on the determined plurality of focus-related values.
- 8.** The system of claim 1, wherein: the area comprises a first area of the sample and a second area of the sample; the plurality of images comprises a first set of images of the first area of the sample and a second set of images of the second area of the sample; the phase diversity analysis comprises a first phase diversity analysis corresponding to the first set of images and a second phase diversity analysis corresponding to the second set of images; the plurality of focus-related values comprises a first set of focus-related values corresponding to the first phase diversity analysis and a second set of focus-related values corresponding to the second phase diversity analysis; the MLE comprises a first MLE of the first set of images and a second MLE of the second set of images; and the focus-corrected image comprises a first focus-corrected image of the first area and a second focus-corrected image of the second area.
- 9.** The system of claim 8, wherein the circuitry is configured to cause the system to further perform: determining, via the first phase diversity analysis: the first set of focus-related values, wherein each focus-related value of the first set of focus-related values is associated with each image of the first set of images; the first MLE of the first set of images; and generating the first focus-corrected image of the first area based on the determined first plurality of focus-related values and the determined first MLE.
- 10.** The system of claim 8, wherein the circuitry is configured to cause the system to further perform: determining, via the second phase diversity analysis: the second set of focus-related values, wherein each focus-related value of the second set of focus-related values is associated with each image of the second set of images; the second MLE of the second set of images; and generating the second focus-corrected image of the second area based on the determined second plurality of focus-related values and the determined second MLE.
- 11.** The system of claim 1, wherein the circuitry is configured to cause the system to further perform: denoising noise in the obtained plurality of images, wherein the noise is modeled as a Poisson distribution; and modeling the denoised noise in the obtained plurality of images on a Gaussian distribution.
- 12.** The system of claim 11, wherein each focus-related value of the plurality of focus-related values is associated with each denoised image of the plurality of images modeled on the Gaussian distribution.
- 13.** A method of improving image quality, the method comprising: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related

values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined MLE.

14. The method of claim 13, wherein each focus-related value of the plurality of focus-related values comprises a defocus value.

15. The method of claim 13, wherein each image of the plurality of images has a different associated focus-related value.

16. The method of claim 13, wherein the plurality of focus-related values comprise a range of focus-related values.

17. The method of claim 13, wherein a range of the plurality of focus-related values corresponds to a range of voltages associated with an objective lens.

18. The method of claim 13, wherein the area is within a field of view of the sample.

19. The method of claim 13, wherein the MLE is determined based on the determined plurality of focus-related values.

20. A non-transitory computer readable medium that stores a set of instructions that is executable by at least one processor of a computing device to cause the computing device to perform operations for improving image quality, the operations comprising: obtaining a plurality of images of an area of a sample; determining via a phase diversity analysis: a plurality of focus-related values, wherein each focus-related value of the plurality of focus-related values is associated with each image of the plurality of images; a maximum likelihood estimate (MLE) of the plurality of images; and generating a focus-corrected image of the area based on the determined plurality of focus-related values and the determined MLE.
